

Michael Malka

Answers AWS:

Section 1: Multiple Choice Questions

1.AWS Core Services

1. C
2. A
3. C

2. IAM & Security

4. A
5. B

3. Networking and Connectivity

6. B
7. A

4. Storage & Database

8. B
9. B

5.AWS Billing &Pricing

10. C
11. A

6. Research-based AWS Questions

12. A landing zone is **a well-architected, multi-account AWS environment that is a starting point from which you can deploy workloads and applications.** It provides a baseline to get started with multi-account architecture, identity and access management, governance, data security, network design, and logging.
13. You use AWS WAF to control how your protected resources respond to HTTP(S) web requests. You do this by defining a web access control list (ACL) and then associating it with one or more web application resources that you want to protect. The associated resources forward incoming requests to AWS WAF for inspection by the web ACL. In your web ACL, you create rules to define traffic patterns to look for in requests and to specify the actions to take on matching requests. The action choices include the following:

Allow the requests to go to the protected resource for processing and response.

Block the requests.

Count the requests.

Run CAPTCHA or challenge checks against requests to verify human users and standard browser use.

14. Snowball is a petabyte-scale data transport solution that uses secure appliances to transfer large amounts of data into and out of the AWS cloud. You use it when you have a large amount of data you need to upload to the cloud.
15. AWS Backup and manual snapshots are very different in their methodologies. AWS Backup is a centralized service for automated backup of multiple AWS services. AWS Backup offers advanced features for cross-region copying, detailed policy management, unified retention strategies, whereas manual snapshots of resources require the management of each resource separately, and they don't offer the same level of automation or pervasive protection as AWS Backup. While manual snapshots can be used for point-in-time backup, AWS Backup is stronger, scalable, and efficient solutions that provide greater security, flexible scheduling, and multi-service support.
16. AWS Shield constantly monitors and inspects traffic through a combination of traffic signature, anomaly algorithms, and other forms of analysis to identify malicious traffic. In case an attack is detected, automatic scaling of resources is performed along with inline deployments that filter malicious traffic, enabling the valid requests to pass through. Most of this happens in less than one second for attacks on the transport layer.
17. VPC Peering creates direct one-to-one connections between VPCs, ideal for smaller setups, while Transit Gateway acts as a central hub for connecting multiple VPCs and on-premises networks at scale. Transit Gateway offers advanced features like dynamic routing, whereas VPC Peering provides simpler routing and lower latency.
18. developers can construct and manage complex, multi-step processes in the cloud with AWS Step Functions, a serverless management tool that helps with workflow automation. It integrates multiple AWS services and offers a visual interface for designing workflows as state machines. By automating decision-making, parallel processing, and error-handling, Step Functions streamlines the creation of distributed systems, lowering application logic complexity and enhancing system reliability in the process. This means instead of handling partial failure cases, retries, and other "non happy cases" as part of your application logic, you can modularize your code into "Steps" and let AWS Step Functions worry about the details.
19. AWS Control Tower is a service that simplifies managing multiple AWS accounts by providing a centralized, automated approach to account governance and compliance. It offers rapid multi-account setup in under 30 minutes, centralized

governance with a single dashboard for monitoring and managing all accounts, enhanced security and compliance with predefined guardrails, automated account provisioning, built-in best practices, consolidated billing, and customizable controls. Control Tower also streamlines account creation and configuration, reduces manual overhead, and allows organizations to apply and enforce specific policies across multiple accounts or units, ensuring flexible governance.

20. AWS Outposts is a key component in hybrid cloud solutions, extending AWS infrastructure to on-premises environments. It allows organizations to run AWS services locally while maintaining consistency with cloud operations. Key benefits include low-latency performance for real-time processing, data residency compliance for industries with strict regulations, and a uniform hybrid experience. Outposts supports AWS services like EC2, EBS, S3, and RDS, making it ideal for applications like modernization, edge computing, industrial automation, and retail analytics. It enables businesses to create flexible, scalable hybrid architectures, optimizing efficiency, reducing costs, and meeting performance and compliance requirements.

21. **EFS:** Shared file storage for multiple EC2 instances, ideal for applications needing a scalable, elastic file system

S3: Object storage for large-scale, unstructured data like backups and media files, with global access and cost-efficient storage classes.

EBS: Block storage for EC2 instances, best for performance-sensitive applications like databases, providing persistent, low-latency storage.

Section 2: Hands-on UI-Based Questions

1. S3 Bucket Configuration

Versioning enabled

michaelbuckettask [Info](#)

Objects | **Metadata** | **Properties** | **Permissions** | **Metrics** | **Management** | **Access Points**

Bucket overview

AWS Region US East (N. Virginia) us-east-1	Amazon Resource Name (ARN) arn:aws:s3:::michaelbuckettask	Creation date February 6, 2025, 23:18:02 (UTC+02:00)
--	---	--

Bucket Versioning [Edit](#)

Versioning is a means of keeping multiple variants of an object in the same bucket. You can use versioning to preserve, retrieve, and restore every version of every object stored in your Amazon S3 bucket. With versioning, you can easily recover from both unintended user actions and application failures. [Learn more](#)

Bucket Versioning
Enabled

Multi-factor authentication (MFA) delete
An additional layer of security that requires multi-factor authentication for changing Bucket Versioning settings and permanently deleting object versions. To modify MFA delete settings, use the AWS CLI, AWS SDK, or the Amazon S3 REST API. [Learn more](#)
Disabled

Tags (0) [Edit](#)

You can use bucket tags to track storage costs and organize buckets. [Learn more](#)

Set a policy to allow only your IAM user to upload objects.

Bucket policy [Edit](#) [Delete](#)

The bucket policy, written in JSON, provides access to the objects stored in the bucket. Bucket policies don't apply to objects owned by other accounts. [Learn more](#)

Public access is blocked because Block Public Access settings are turned on for this bucket
To determine which settings are turned on, check your Block Public Access settings for this bucket. [Learn more about using Amazon S3 Block Public Access](#)

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Principal": {
        "AWS": "arn:aws:iam::504949722475:user/jlb-user"
      },
      "Action": "s3:PutObject",
      "Resource": "arn:aws:s3:::michaelbuckettask/*"
    }
  ]
}
```

[Copy](#)

2. Launch an EC2 Instance

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 address
michael-ec2-instance	i-0fb0c8766d77b89fd	Running	t2.micro	Initializing	View alarms +	us-east-1a	ec2-52-90-106-5.comp...	52.90.106.5

Inbound rules						
Name	Security group rule ID	Port range	Protocol	Source	Security groups	Description
-	sgr-00829e205a2efd15e	22	TCP	0.0.0.0/0	launch-wizard-36	-
-	sgr-03410e1f95e5ebd6a	80	TCP	0.0.0.0/0	launch-wizard-36	-

3. Configure an IAM User with S3 Access

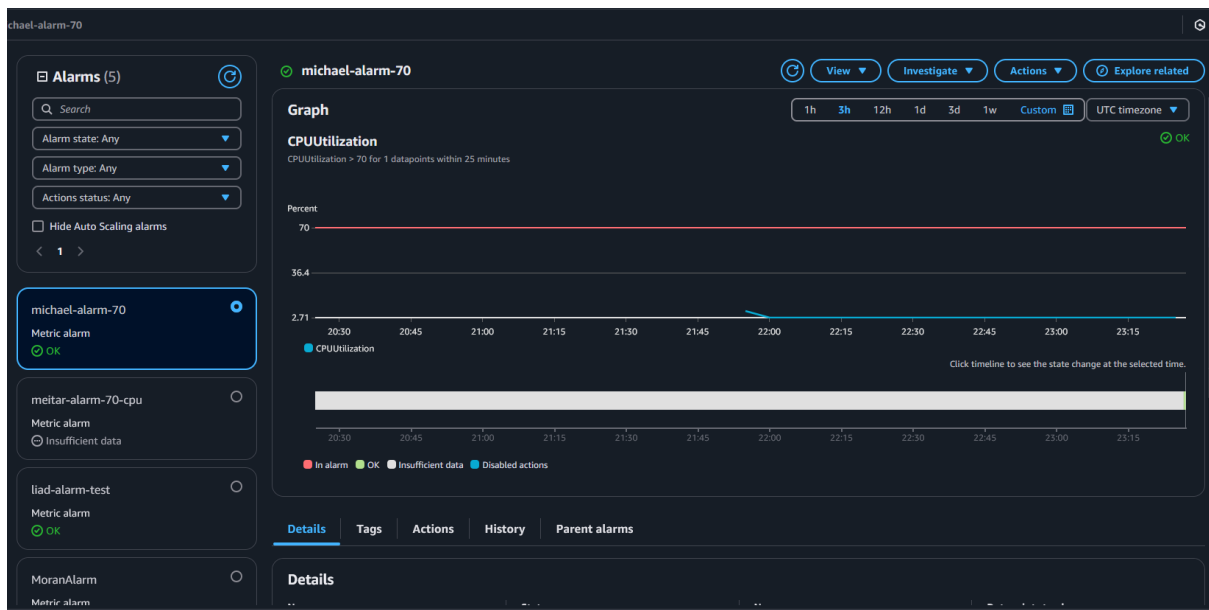
Summary		
ARN arn:aws:iam::504949722475:user/michael-user	Console access Enabled without MFA	Access key 1 Create access key
Created February 07, 2025, 10:46 (UTC+02:00)	Last console sign-in Today	

Permissions policies (1)
Permissions are defined by policies attached to the user directly or through groups.
Filter by Type All types
Policy name Type Attached via
michael-policy-task-s3 Customer managed Directly

Add photo to the bucket to my user that I created (Michael-user) have permission to

Upload			
Add the files and folders you want to upload to S3. To upload a file larger than 160GB, use the AWS CLI, AWS SDKs or Amazon S3 REST API. Learn more			
Drag and drop files and folders you want to upload here, or choose Add files or Add folder .			
Files and folders (1 total, 457.6 KB)			
All files and folders in this table will be uploaded.			
Find by name			
Name	Folder	Type	Size
NanoShark.jpg	-	image/jpeg	457.6 KB

4. Set Up a CloudWatch Alarm



Simple Notification Service

Subscription confirmed!

You have successfully subscribed.

Your subscription's id is:

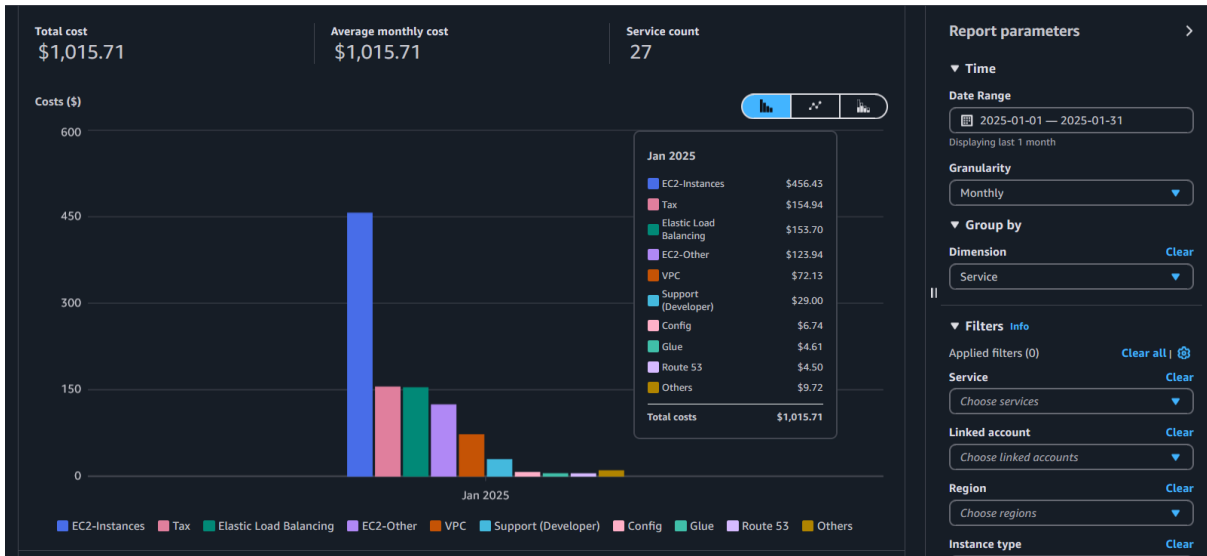
arn:aws:sns:us-east-1:504949722475:CPU-70OverLoad:6356dd52-ddd9-4918-8843-5d6a7a1c8625

If it was not your intention to subscribe, [click here to unsubscribe](#).

Description creating the alarm:

I set a threshold for triggering if CPU usage exceeds 70% for 5 minutes and specify the evaluation period. Link an SNS topic with an email subscription for notifications.

5. Identify AWS Billing Costs



The total cost \$1,015.71 across 27 services, with EC2 Instances as the highest expense, followed by Taxes, Elastic Load Balancing, and VPC. Costs are grouped by service without additional filters. Potential optimizations include right-sizing EC2 instances, reserved instances, and reviewing networking costs.

Section 3: Hands-on advanced

1. Deploy an Auto Scaling Group with a Single EC2 Instance

michael-template (lt-01c5b25df5d6bfad1) Actions Delete template

Launch template details

Launch template ID
lt-01c5b25df5d6bfad1

Launch template name
michael-template

Default version
1

Owner
arn:aws:iam::504949722475:user/jb-user

Details **Versions** **Template tags**

Launch template version details Actions Delete template version

Version
1 (Default)

Description
-

Date created
2025-02-07T10:07:49.000Z

Created by
arn:aws:iam::504949722475:user/jb-user

Instance details **Storage** **Resource tags** **Network interfaces** **Advanced details**

AMI ID
ami-04681163a08179f28

Instance type
t2.micro

Availability Zone
-

Key pair name
michaelKeyForTask

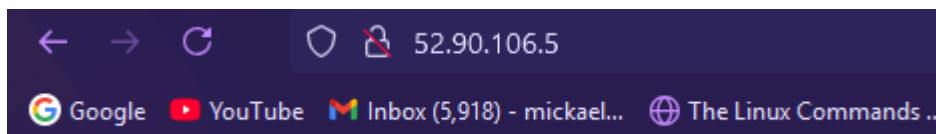
Security groups
-

Security group IDs
sg-0bdf2e671d92fabca


```
michael@MichaelPC:~$ telnet 52.90.106.5 22
Trying 52.90.106.5...
Connected to 52.90.106.5.
Escape character is '^]'.
SSH-2.0-OpenSSH_8.7
```

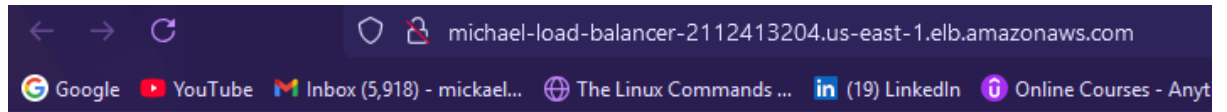
```
● nginx.service - The nginx HTTP and reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: disabled)
   Active: active (running) since Fri 2025-02-07 11:04:35 UTC; 30s ago
     Main PID: 51990 (nginx)
       Tasks: 2 (limit: 1111)
      Memory: 2.5M
         CPU: 43ms
    CGroup: /system.slice/nginx.service
            └─51990 "nginx: master process /usr/sbin/nginx"
              └─51991 "nginx: worker process"
```

```
[ec2-user@ip-172-31-87-165 ~]$ curl http://localhost:80
<h1>Welcome to AWS Auto Scaling</h1>
```



Welcome to AWS Auto Scaling

3. Access the Web Page via the Load Balancer



Welcome to AWS Auto Scaling

4. IAM User Setup for S3 Access

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "S3BucketAccess",
      "Effect": "Allow",
      "Action": [
        "s3:ListAllMyBuckets",
        "s3:ListBucket",
        "s3:GetBucketLocation",
        "s3:ListMultipartUploadParts",
        "s3:ListBucketMultipartUploads"
      ],
      "Resource": "arn:aws:s3:::*"
    },
    {
      "Sid": "SpecificBucketAccess",
      "Effect": "Allow",
      "Action": [
        "s3:PutObject",
        "s3:GetObject",
        "s3:DeleteObject"
      ],
      "Resource": "arn:aws:s3:::michaelbuckettask/*"
    }
  ]
}
```

[Alt+S]

VPC Elastic Beanstalk CloudFormation Route 53 CloudWatch Elastic Container Registry EC2 IAM S3

United States (N. Virginia) michael-user @ jb-labs

Account snapshot - updated every 24 hours All AWS Regions View Storage Lens dashboard

Storage lens provides visibility into storage usage and activity trends. Metrics don't include directory buckets. [Learn more](#)

General purpose buckets Directory buckets

General purpose buckets (78) Info All AWS Regions

Buckets are containers for data stored in S3.

Find buckets by name

Name	AWS Region	IAM Access Analyzer	Creation date
albert-website-765	US East (N. Virginia) us-east-1	View analyzer for us-east-1	February 4, 2025, 19:55:18 (UTC+02:00)
almog-devops-s3	US East (N. Virginia) us-east-1	View analyzer for us-east-1	February 7, 2025, 09:14:19 (UTC+02:00)
amitbe-bucket	US East (N. Virginia) us-east-1	View analyzer for us-east-1	February 6, 2025, 21:47:18 (UTC+02:00)
assaf-bkt-02-06	US East (N. Virginia) us-east-1	View analyzer for us-east-1	June 2, 2024, 20:02:02 (UTC+03:00)
aws-cloudtrail-logs-504949722475-084cc1f8	US West (Oregon) us-west-2	View analyzer for us-west-2	July 11, 2023, 10:02:03 (UTC+03:00)
aws-cloudtrail-logs-504949722475-969fa94d	US West (Oregon) us-west-2	View analyzer for us-west-2	July 11, 2023, 10:03:29 (UTC+03:00)
aws-glue-assets-504949722475-us-east-1	US East (N. Virginia) us-east-1	View analyzer for us-east-1	January 22, 2025, 08:38:49 (UTC+02:00)
aws-logs-504949722475-us-east-1	US East (N. Virginia) us-east-1	View analyzer for us-east-1	January 7, 2025, 10:34:19 (UTC+02:00)
bennytfr-prod-jb-us-east-1	US East (N. Virginia) us-east-1	View analyzer for us-east-1	January 27, 2025, 20:31:34 (UTC+02:00)
beri-s3-sample-01	US East (N. Virginia) us-east-1	View analyzer for us-east-1	September 3, 2023, 20:05:33 (UTC+03:00)
cf-storageclusters	Europe (Ireland) eu-west-1	View analyzer for eu-west-1	December 1, 2022, 15:44:42 (UTC+02:00)
cf-templates-1xm5o5vx382os-eu-central-1	Europe (Frankfurt) eu-central-1	View analyzer for eu-central-1	January 28, 2024, 20:04:43 (UTC+02:00)
cf-templates-1xm5o5vx382os-us-east-1	US East (N. Virginia) us-east-1	View analyzer for us-east-1	August 9, 2023, 19:19:29 (UTC+03:00)
cf-templates-1xm5o5vx382os-us-west-2	US West (Oregon) us-west-2	View analyzer for us-west-2	January 28, 2024, 18:53:08 (UTC+02:00)

[Alt+S]

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General purpose buckets Directory buckets

General purpose buckets (78) Info All AWS Regions

Buckets are containers for data stored in S3.

mic 1 match

Name	AWS Region	IAM Access Analyzer	Creation date
michaelbuckettask	US East (N. Virginia) us-east-1	View analyzer for us-east-1	February 6, 2025, 23:18:02 (UTC+02:00)

5. Create a CloudWatch Alarm for CPU Usage

Already done in section 2 question 4